Garrett Hagen

Software Engineer

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EXPERIENCE

Apple Inc — Apple TV App
iOS Software Engineer · *July 2022 – Present*

- Delivered Apple TV App functionality for Apple Intelligence–powered Siri through cross-org collaboration with the Siri and Intents frameworks teams.
- Own the TV Home Screen widget, shipping new features and improvements across Apple platforms.
- Built client support for TV App features including condensed and full sports replays and partner-app integration features.
- Own the client instrumentation that measures app launch and page-load performance and reports UI events across the TV App, and refactored it to adopt standardized internal frameworks.
- Led the UI and data-modeling layer for an upcoming iOS feature.

Dexcom Inc — Continuous Glucose Monitoring Systems

iOS Engineer, New Markets Team · Aug 2021 – July 2022

- Owned transmitter integration (G6 and G7) for the hospital CGM iOS app, maintaining and integrating Dexcom's existing Bluetooth LE communication frameworks.
- Built and improved areas of the patient onboarding flow.

QA Engineer, Transmitter Team · Aug 2020 – Aug 2021

- Wrote automated tests exercising real-world usage scenarios between CGM transmitters and mobile apps over Bluetooth LE.
- Built an Appium suite automating typical usage scenarios to reproduce and analyze BLE connection drops, plus tooling to inspect the captured data.

Gener-8 — FDA-Approved CPM Self-Rehabilitation Devices
Software Development & Hardware Design · *2016 – Present · Part-time*

Second-generation system · in progress

- Redesigned the machine hardware around a Raspberry Pi system and developed multi-threaded firmware in Python.
- Added real-time calibration offsets, improved LED feedback, and an extensive Bluetooth LE feature set: headless Wi-Fi configuration, wireless firmware updates, motor-speed control, sensor-state reading, rep counting, and task-progress updates.
- Migrating the app's client logic to defined Firebase Cloud Functions endpoints to centralize business logic and thin the client.
- Moving the codebase from CocoaPods to Swift Package Manager, splitting shared code into reusable Gener-8 packages consumed by both the factory-setup and production apps.
- Wrote Python admin tools and manufacturing software for the platform, spanning system database administration and Raspberry Pi device provisioning.
- Refactored the Gener-8 website using agentic, AI-assisted engineering workflows.
- Built an agentic development harness (Ralph loop) coordinating multiple Claude agents across the BLE peripheral and a paired iOS device — one SSHing into the Gener-8 to tail logs and run commands, another driving and monitoring the iOS app — enabling automated, remote development and debugging.

First-generation system

- Designed the circuitry and embedded software for an on-device angle-calculation system that streams real-time IMU measurements over Bluetooth LE.
- Shipped the Gener-8 iOS app — machine communication, calibration, patient workout-data capture and storage, patient–therapist messaging, and rehab visualization and feedback.
- Owned the full technical stack, from circuit design and manufacturing to app development and App Store deployment.

Carnegie Robotics — Advanced Robotics Sensors & Platforms
Software Co-op · *2018 – 2019*

- Created and maintained a sizeable C++ framework that reduced boilerplate for projects using stereo-camera functionality.
- Built a C++ application assessing extrinsic/intrinsic calibration quality for MultiSense cameras using OpenCV, Ceres-Solver, Boost, and GTest.
- Developed an interactive Qt/OpenCV C++ GUI for concurrent grayscale, color, and disparity streaming across all MultiSense camera models.
- Contributed embedded C firmware and Python tooling for MultiSense PCB verification, ROS/OpenCV data-collection utilities, and Docker/CMake build and deployment improvements.

PATENTS

Durabat® — baseball training device; designed and patented, used by professional players.

SKILLS

Languages: Swift, Python, Objective-C, TypeScript, Java, C++

iOS / Frameworks: CoreBluetooth (BLE), SwiftUI, UIKit, WidgetKit, AppIntents, Firebase, Xcode

Tools: Git, CMake, Appium, JUnit, AutoDesk Inventor, Claude Code

Platforms: iOS, tvOS, macOS, visionOS, Linux

EDUCATION

University of Pittsburgh
BS, Computer Engineering · 2016–2020

- Major GPA 3.82 · Cumulative 3.55
- Engineering Dean's Honor List (2017–2020)

PROJECTS + OPEN SOURCE

wpa-pyfi *2021*

- Open-source Python package that adapts and extends an existing framework to programmatically manage Raspberry Pi Wi-Fi connections and settings from the CLI or as a library. Published on PyPI.

OmniBot *2020*

- Omnidirectional robot built with a student-led team for a senior capstone. Developed the iOS app and Bluetooth LE control protocol (joystick, autopilot, and gesture control), training an on-device image-recognition model on hand gestures that drove the robot from the phone's camera feed.

Biometric Lock *2019*

- Swift iOS app communicating over Bluetooth LE with an Arduino-driven fingerprint scanner and solenoid lock. Designed all software and circuitry and rendered a custom enclosure mounted in the door frame.

Industrial Arduino *2017*

- Custom PCB shield designed in Altium Designer for a waterproof, industrial Arduino enclosure.

VOLUNTEERING

HCEF Instructor *2019–2020*

- Taught basic computer science to middle-school students experiencing homelessness.